Chromatic Accessibility in Latin American University Websites: A Large-Scale Survey of WCAG Contrast Compliance

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Abstract

Web accessibility audits at scale have historically concentrated on North American and European institutions, leaving higher education in Latin America largely unexamined. We crawled the homepages of 464 Latin American university websites across 17 countries and successfully extracted computed text/background colour pairs from 259 of them (41,360 pairs in total). Each pair was converted from sRGB to CIELAB and evaluated against the WCAG 2.2 contrast thresholds, with the axe-core accessibility engine as an automated cross-check ($r = 0.347$). We find that 21.6% of all colour pairs fail the AA threshold, and 93.5% of sites contain at least one critical failure. A pair-level chi-square test suggests compliance differs by country ($\chi^2 = 1625.04$, $df = 16$, $p < 0.001$, Cramér's $V = 0.198$), but because colour pairs are clustered within sites rather than independent, we treat this as exploratory: a site-level test on the true unit of analysis (256 sites, countries with $\geq$5 sites) is not significant ($\chi^2 = 22.98$, $df = 15$, $p = 0.085$). A within-site typography effect (large text failing less often) does not survive site-clustered sandwich standard errors ($p = 0.473$), although continuous font size in pixels retains a small association with failure ($p = 0.027$). Unsupervised clustering of the failing pairs in the full CIELAB $L^*a^*b^*$ space supports the initial hypothesis: near-neutral, low-chroma clusters consistent with the "light grey on white" pattern account for 80.0% of all failing pairs, with four smaller chromatic clusters spanning the remaining 20.0%.

Keywords: Colour contrast, Chromatic accessibility, WCAG 2.2, CIELAB colour space, Web accessibility, Information design.

1 Introduction

Web accessibility is frequently framed as a compliance checkbox rather than as a problem of information design: which visual choices allow a message to be perceived by the range of readers who will actually encounter it. Colour contrast between text and its background is one of the most consequential of those choices, because unlike many other accessibility barriers it is silent: a page can look complete to a sighted designer working under ideal lighting on a calibrated monitor while remaining functionally illegible to a large fraction of visitors.

Large-scale, empirical audits of contrast compliance exist, most notably WebAIM's annual scan of the top one million global home pages (WebAIM, 2026), which finds low-contrast text on 83.9% of pages and identifies it as the single most common accessibility failure detected, unchanged in rank for seven consecutive years. Such global scans report an aggregate world figure, though, and are not designed to, and do not, break results out for Latin American higher education specifically. Latin American higher education, more than 460 institutions across 19 countries, has not, to our knowledge, been the subject of a comparable regional survey. This is a meaningful gap: university websites are a primary public-facing channel for admissions, financial aid, and academic information, and design conventions, content management systems, and institutional resourcing in the region differ enough from those studied elsewhere that findings cannot simply be assumed to transfer.

This paper reports a large-scale survey of chromatic accessibility on the homepages of Latin American universities. Rather than relying on a small hand-picked sample, we build a stratified list of institutions from an open, community-maintained directory, crawl each homepage with a real browser engine to capture computed (post-CSS) colour pairs, and classify every pair against the WCAG 2.2 (World Wide Web Consortium (W3C), 2023) contrast thresholds. We ask two questions. First, at what rate do these institutions fail contrast requirements, and does that rate vary systematically by country? Second, what visual pattern, if any, characterises the colours involved in a failure? The second question matters because the answer determines what kind of intervention would actually help: a narrow pattern (e.g., light grey body text) suggests a simple style-guide fix, while a diffuse one suggests a more structural design-literacy problem.

The contributions of this work are: (1) the first large-scale regional survey of chromatic accessibility for Latin American university homepages, built from a stratified sample of 464 institutions across 17 countries; (2) a site-level, clustering-aware statistical treatment of the country question that distinguishes a descriptive pair-level pattern from a confirmed site-level effect; (3) an unsupervised characterisation, in the full CIELAB colour space, of the visual patterns underlying contrast failures in this region, showing that two low-chroma, near-neutral clusters (a near-white and a mid-grey pattern) jointly dominate, with four smaller chromatic clusters accounting for the remainder.

2 Related Work

Contrast and legibility research inside the graphic design literature has largely focused on typography and reading performance rather than on regulatory thresholds. Weingerl, Nedeljković & Pušnik (2022) studied the visibility and legibility of short words under controlled experimental conditions, establishing that legibility judgments depend jointly on typographic and perceptual factors rather than on any single variable in isolation, a finding that motivates our decision to treat contrast as one factor among several (font size, weight) rather than in isolation. Punsongserm & Suvakunta (2025) conducted a technical review of typeface choice, type size, and colour contrast in government mobile applications, and, closer to our own setting, found systematic WCAG non-compliance in public-sector digital products, a pattern we replicate, in a different modality and region, for public university websites. Ofosu-Asare (2024) examined usability and design evaluation techniques for online learning environments, noting that accessibility in educational digital products is usually assessed qualitatively or on small samples rather than through a large-scale automated survey, which is the gap this paper addresses directly for the chromatic dimension. Chromatic reproduction in printed media (including the drying and ink formulation properties that govern gamut and delta-E) has been studied within this journal's tradition (Saad et al., 2021); the digital counterpart of that problem, namely whether the sRGB palette chosen for web interfaces falls within the perceptually accessible gamut, is the chromatic problem this paper addresses.

The conceptual foundation of our study rests on the universal web accessibility argument advanced by Henry, Abou-Zahra & Brewer (2014), who established that accessibility is not a special-case concern for users with disabilities but a universal quality requirement: low contrast degrades legibility for all users under adverse ambient conditions (sunlight, low-quality screens, ageing displays), making compliance with WCAG contrast thresholds a baseline quality requirement rather than a minority accommodation. Rello & Baeza-Yates (2013) further demonstrated that typographic and chromatic choices jointly affect reading speed and comprehension for users with dyslexia, a population not typically captured by automated accessibility audits focused on WCAG pass/fail, reinforcing the argument that a perceptual analysis in continuous CIELAB space, rather than a binary threshold test, provides richer information about the actual user experience.

The closest direct precedent in the Latin American higher-education domain is the body of work by Acosta-Vargas and collaborators, who have conducted the most systematic prior surveys of university website accessibility in the region. Their 2018 study (Acosta-Vargas, Acosta & Lujan-Mora, 2018) evaluated 348 Latin American university websites using the WAVE automated accessibility engine, cataloguing a broad range of WCAG 2.0 failures and confirming that colour contrast is among the most frequently occurring ones; it is the benchmark against which our 93.5% per-site failure rate and our 21.6% pair-level failure rate should be positioned. A subsequent heuristic method for low-vision accessibility (Acosta-Vargas, Salvador-Ullauri & Lujan-Mora, 2019) extended this to a perceptual dimension, arguing that WCAG's relative-luminance threshold is a necessary but not perceptually sufficient criterion, a theoretical position that directly motivates our decision to cluster failing pairs in the full CIELAB $L^*a^*b^*$ space rather than stopping at the WCAG ratio alone. Their 2020 dataset (Acosta-Vargas, González & Luján-Mora, 2020) makes the evaluation data for 368 pages publicly available, enabling direct comparison with

automated-tool outputs; our study is methodologically complementary in that we use a different pipeline (computed-style extraction via Playwright rather than a WAVE rule-engine scan) and restrict the analysis to the chromatic dimension specifically, extracting every text/background colour pair in CIELAB coordinates rather than returning a binary pass/fail per element, which allows us to characterise the failure patterns in continuous colour space. More recent work by the same group confirms that the contrast failure pattern extends beyond the university sector: Acosta-Vargas et al. (2022) found contrast errors to be the dominant accessibility barrier in 50 top-ranked e-commerce websites (83.1% of failures classified as "perceivable" violations), and a complementary mobile-app study (Acosta-Vargas et al., 2021) reported that 86% of Ecuadorian Android applications in Google Play failed WCAG contrast criteria, situating our university-website result within a broader pattern of contrast non-compliance across the Latin American digital product ecosystem.

3 Methodology

3.1 Site selection

We built the candidate list from the *Hipo/university-domains-list* open dataset, filtered to the 19 countries of Latin America, and drew a stratified random sample across countries (capped per country to avoid Brazil and Mexico dominating the list). The final list contained 464 domains across 17 countries; the source dataset contained no entries for Bolivia or Venezuela, which are consequently absent from this study (see Limitations).

3.2 Crawling

Each homepage was visited with a headless Chromium browser (Playwright), identifying our crawler with a descriptive user agent and honoring `robots.txt` (fetched with the same identifying user agent, since fetching it with a generic client triggers 403 responses on several sites and would otherwise cause valid, crawlable sites to be excluded by mistake). Requests were rate-limited to one every two seconds. Of the 464 domains, 278 (59.9%) were successfully rendered; the remainder were unreachable, timed out, or were excluded by `robots.txt` (see Table 1).

3.3 Colour pair extraction

For each of the 278 reachable sites we revisited the live page a second time and, via `getComputedStyle()`, extracted every visible text node's computed text colour together with its *effective* background colour, resolved by walking up the DOM ancestor chain until a non-transparent background is found, since a text element's immediate parent is frequently transparent. Elements with zero width/height, `display:none`, `visibility:hidden`, or zero opacity were discarded. This step requires a live browser rather than the raw HTML saved during crawling, because a serialised DOM does not inline externally linked CSS; computing style from the saved markup alone would silently fall back to browser defaults. This yielded 41,360 colour pairs from 259 sites (19 sites returned no usable pairs, typically due to navigation during evaluation).

3.4 Contrast calculation and cross-validation

Each pair's WCAG relative luminance and contrast ratio were computed directly from sRGB, and each pair was independently converted from sRGB to CIELAB (International Commission on Illumination (CIE), 1976) (D_{65} illuminant) for the clustering analysis. As a cross-check, we additionally ran the `axe-core` accessibility engine (the library underlying most commercial accessibility linters) against the same live page and recorded its colour-contrast violation and pass counts.

3.5 Classification and aggregation

Following WCAG 2.2, each pair was classified as *fail*, *AA*, or *AAA* using thresholds that depend on text size: 3:1 (AA) / 4.5:1 (AAA) for large text ($\geq$24px, or $\geq$18.66px bold), and 4.5:1 (AA) / 7:1 (AAA) otherwise. Results were aggregated per site (median ratio, % of pairs failing AA, whether the site has at least one critical failure).

3.6 Statistical analysis

We tested independence between country and AA compliance with a χ^2 test (continuity-corrected, restricted to countries having ≥ 30 pairs) and we report Cramér's V as the effect size. This test, like any test run on the 41,360-pair table, treats colour pairs as independent observations, but pairs from the same site share a common design system and are not independent draws, a form of pseudo-replication that inflates the effective sample size and can make a null or weak site-level effect appear significant at the pair level. To address this directly rather than only flag it, we repeat the country comparison at the site level, the unit at which country actually varies: a χ^2 test on each site's binary critical-failure status, and a Kruskal-Wallis test on each site's continuous %AA-failure, both restricted to the 16 countries with $\geq$5 sites (256 of 259 sites). We fit a logistic regression of pair-level failure on country, font size, and whether the text qualifies as "large," and repeat it with site-clustered sandwich standard errors (generalised estimating equations, independence working correlation, clustered by site) to check which effects survive proper clustering rather than only the country effect. We clustered the failing pairs' text-colour coordinates in the full, standardised CIELAB $L^*a^*b^*$ space with k-means, rather than the a^*b^* chroma plane alone, since WCAG failure is a function of luminance and a chroma-only clustering would place every achromatic colour at the origin regardless of lightness; we searched $k \in \{2, \ldots, 20\}$ for a silhouette-optimal solution before fixing $k = 6$ for interpretability (Results).

4 Results

Table 1: Crawl outcomes for the 464 sampled domains.

Outcome	Count
Successfully rendered	278
Unreachable / timed out / other error	167
Excluded by `robots.txt`	19

Across the 41,360 extracted colour pairs, 21.6% fail the applicable WCAG AA threshold, 16.6% meet AA but not AAA, and 61.8% meet AAA (Figure 1). At the site level, 93.5% of the 259 successfully analysed homepages contain at least one critical (failing) pair, and the median site has 19.9% of its pairs failing AA, so failures are not concentrated in a handful of outlier institutions; they are the norm. This per-page failure rate is higher than WebAIM's 83.9% figure for the global top one million home pages (WebAIM, 2026), though the two figures come from different detection pipelines applied to different populations and should be read as an order-of-magnitude reference rather than a controlled comparison (Limitations).

Table 2: Descriptive statistics by country (16 countries with $\geq$5 sites; 256 of 259 sites).

Country	Sites	Median % AA-fail	% sites, $\geq$1 critical fail
Argentina	27	22.0	96.3
Chile	27	15.0	96.3
Brazil	26	12.4	92.3
Colombia	26	19.0	100.0
Ecuador	24	20.1	91.7
Peru	21	25.4	81.0
Costa Rica	19	24.0	94.7
Mexico	15	9.0	73.3
Dominican Republic	15	33.6	100.0
El Salvador	13	15.5	100.0
Guatemala	10	22.3	100.0
Nicaragua	9	33.3	88.9
Panama	7	14.3	100.0
Cuba	6	31.6	100.0
Honduras	6	22.9	83.3
Paraguay	5	15.6	100.0

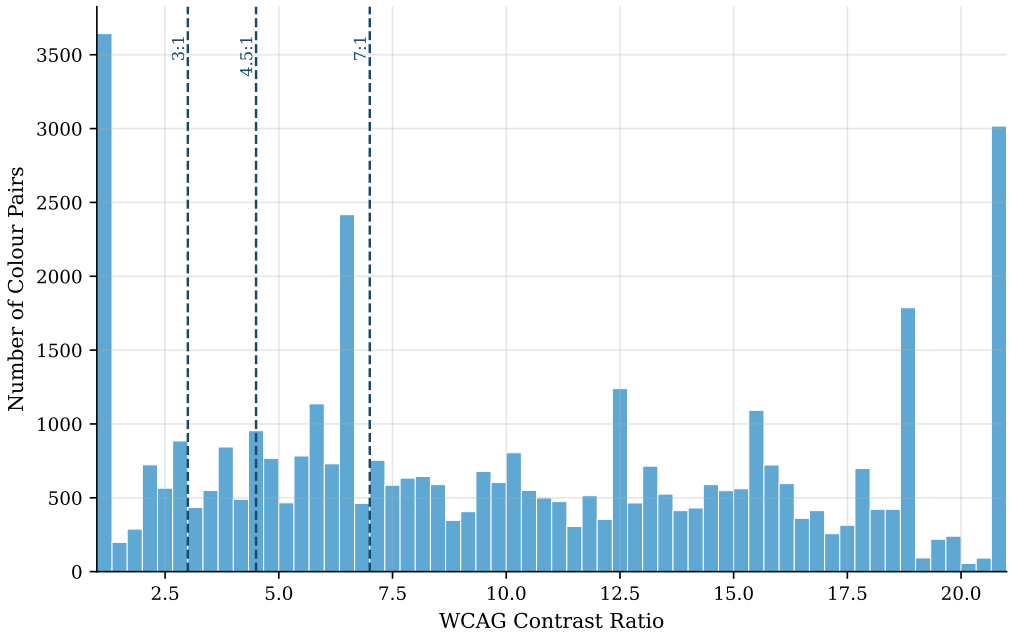


Figure 1: Distribution of WCAG contrast ratios across the 41,360 extracted colour pairs, with the AA and AAA thresholds marked.

Table 2 gives descriptive statistics by country before turning to the significance question. The pair-level test suggests compliance varies by country ($\chi^2 = 1625.04$, $df = 16$, $p < 0.001$, Cramér's $V = 0.198$; Figure 2), and the pair-level logistic regression finds most country coefficients individually significant (e.g., Paraguay: $\beta = -2.23$, $p < 0.001$; Cuba: $\beta = +0.57$, $p < 0.001$). We do not take this at face value: because pairs are clustered within sites, the pair-level test's effective sample size (41,360) overstates independent information, and its p-values should be read as optimistic. The site-level test, restricted to the 256 sites in the 16 countries with $\geq$5 sites, is *not* significant either on critical-failure status ($\chi^2 = 22.98$, $df = 15$, $p = 0.085$) or on continuous %AA-failure (Kruskal-Wallis $H = 21.53$, $p = 0.121$). Because the site-level critical-failure rate is high (most countries have few or no compliant sites), several cells of the country $\times$ failure contingency table have an expected count well below the conventional rule-of-thumb minimum of 5 for the chi-squared approximation to be reliable (minimum expected count = 0.33); as a sensitivity check we therefore also ran a 10,000-iteration Monte Carlo permutation test (shuffling critical-failure status across sites within the same country distribution and recomputing χ^2 each time), which agrees closely with the asymptotic result ($p = 0.081$), so this is not an artefact of the chi-squared approximation breaking down. We therefore report the country effect as a real but unresolved pattern (visible and directionally consistent at the pair level, but not confirmed at the site level with the sample sizes available here) rather than as a confirmed finding, and we do not treat the specific per-country coefficients above as reliable individually. We initially expected typography to be on firmer ground, reasoning that because "large text" varies within a site (a single page mixes body and heading sizes) its effect would not be vulnerable to the same between-site clustering concern. This expectation does not survive a direct check: refitting the logistic regression with site-clustered sandwich standard errors (a generalised estimating equations specification, independence working correlation, clustered by site) rather than pooled naive ones leaves the large-text coefficient unchanged in size ($\beta = -0.10$) but no longer significant ($p = 0.473$, versus $p = 0.022$ pooled), so we do not treat it as a reliable finding. Raw font size in pixels fares better: it retains a significant, though far more modest, *positive* association with failure under the same clustered specification ($\beta = +0.0108$, $p = 0.027$, versus $p < 0.001$ pooled). Of the two typography variables tested, only continuous font size in pixels survives cluster-adjusted inference, and only marginally; the categorical large-text effect does not.

We clustered the 8,925 failing pairs' text-colour coordinates in the full, standardised CIELAB $L^*a^*b^*$ space with k-means at $k = 6$ (silhouette = 0.796); we cluster on all three coordinates, not a^*b^* alone, since WCAG failure is governed by luminance and a chroma-only projection would collapse every achromatic colour (black, grey, white) onto the origin regardless of its lightness, which would settle the composition question by construction rather than measure it. We searched silhouette over $k \in \{2, \ldots, 20\}$ as a

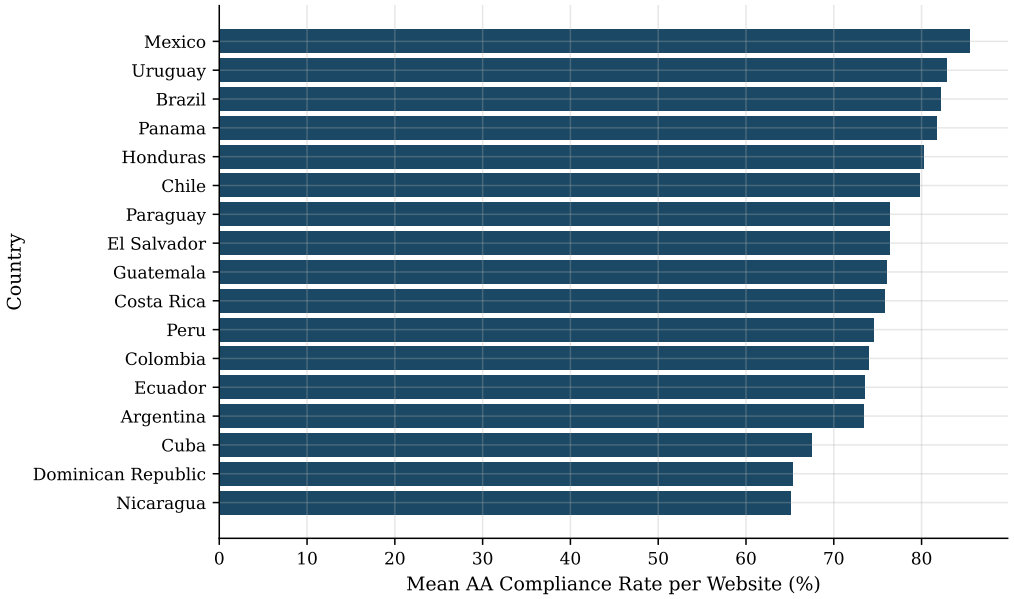


Figure 2: Mean AA compliance rate by country at the website level ($n \geq 3$ sites per country, 17 countries including Uruguay, 259 total sites; descriptive; see main text for why the associated significance test is not treated as confirmatory). Note that Table 2 restricts descriptive statistics to the 16 countries with $n \geq 5$ sites (256 sites) for statistical robustness.

sensitivity check rather than selecting k by silhouette maximisation: the score does not have a clear interior maximum over that range, so we fix $k = 6$ for interpretability. The resulting partition separates the low-chroma pairs into two clusters at different lightness levels rather than merging them: a near-white cluster (mean $L^* = 99.5$, chroma ≈ 0.2, 60.1% of failing pairs) and a mid-grey cluster (mean $L^* = 51.7$, chroma ≈ 5.0, 19.9%), together 80.0% of all failing pairs, with four markedly smaller chromatic clusters (4.1% to 6.1% of pairs each, chroma 41.6–78.4) spanning red, blue, yellow-green, and green regions of the plane (Figure 3). The near-white cluster's lightness ($L^* = 99.5$, indistinguishable from pure white) is consistent with the background-image detection artefact discussed in Limitations: at least part of this cluster is likely mislabelled text-over-image rather than genuinely near-white text. Agreement between our contrast-ratio-based failure count and `axe-core`'s independent colour-contrast violation count is positive but moderate ($r = 0.347$ across sites), which is expected given the two tools differ in which elements they consider evaluable (axe-core additionally skips certain non-text and gradient cases that our extraction still captures).

Table 3: The 10 most frequent failing colour pairs, with median WCAG contrast ratio and median CIELAB ΔE (CIE76). ΔE is evaluated using Euclidean distance in CIELAB space (CIE76), which provides an intuitive, geometrically transparent distance metric directly aligned with the k-means feature space, whereas CIEDE2000 incorporates complex perceptual weighting factors unnecessary for standard Euclidean clustering.

Text (hex)	Background (hex)	n pairs	Ratio (median)	ΔE (median)
#FFFFFF	#FFFFFF	2,379	1.00	0.0
#777777	#FFFFFF	186	4.48	50.0
#9F6B6B	#F2F2F2	172	3.92	50.2
#E33B3B	#FFFFFF	160	4.24	89.4
#FFFFFF	#EDEDED	137	1.17	6.3
#BFAC51	#FFFFFF	124	2.27	57.0
#999999	#FFFFFF	116	2.85	36.8
#00BCD4	#FFFFFF	87	2.30	48.8
#FFFFFF	#F2F2F2	81	1.12	4.5
#959595	#FFFFFF	76	3.00	38.3

Table 3 shows that the single most frequent failing pair, by a wide margin, is white text on a white background (2,379 occurrences, $\Delta E = 0$). We cannot fully distinguish two explanations from the extracted data: text genuinely styled to match its background (a fully invisible pair), and text rendered over a photographic or gradient `background-image`, which our background-colour resolution (Methodology, Limitations) would also record as white-on-white, since it only inspects `background-color` up the DOM ancestor chain. Both explanations describe real accessibility failures, but only the first is a literal invisible-text pattern; the practical implication (this is the single largest identifiable failure signature in the dataset) holds either way.

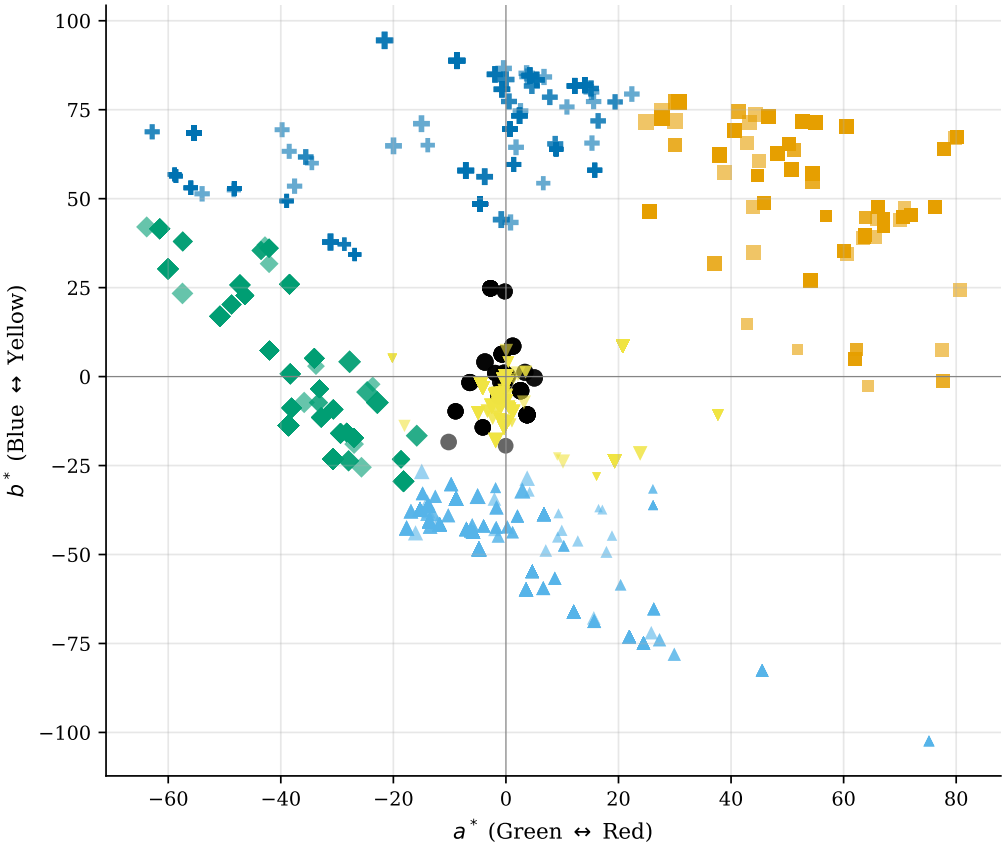


Figure 3: Failing pairs' text-colour coordinates projected onto the CIELAB a^*b^* plane, coloured by k-means cluster ($k = 6$, computed in the full $L^*a^*b^*$ space); point size encodes L^* (lightness).

5 Discussion

That most Latin American university homepages fail WCAG contrast requirements mirrors findings from other regions and is, on its own, unsurprising. Our working hypothesis, drawn from common assumptions in the accessibility-audit literature, was that failures would concentrate in familiar patterns such as light grey text on white, text over unmasked photographic backgrounds, and unverified interaction states (hover/focus/visited). The clustering result supports the low-chroma part of that hypothesis directly: near-neutral clusters consistent with the "light grey on white" pattern account for 80.0% of failing pairs, split by our full-$L^*a^*b^*$ clustering into a near-white cluster ($L^* = 99.5$, 60.1%) and a mid-grey cluster ($L^* = 51.7$, 19.9%), not merely one of several comparable clusters. The near-white cluster's dominance is likely inflated by the background-image blind spot discussed in Limitations, since text over an unmasked photograph would also resolve as a near-white failing pair under our extraction method rather than as a distinct chromatic cluster; the mid-grey cluster is not subject to that concern, since $L^* = 51.7$ is far from the near-white artefact region. Even allowing for that overlap in the near-white cluster, the remaining chromatic clusters, saturated brand colours used directly as text colour, are each individually small (4.1% to 6.1% of pairs), so the practical intervention implied by our results is closer to the field's existing assumption than we initially expected: a style-guide fix targeting low-contrast neutral and grey text

would address the large majority of observed failures, with saturated brand-colour misuse a secondary, smaller pattern.

For the institutions themselves, a style guide targeting low-contrast neutral (grey-on-white) text would directly address the large majority of the failures we observe. The smaller chromatic clusters are still worth a secondary mention: a plausible, though untested, explanation for that residual is institutional brand palettes (a university's primary colour used directly as a text colour, regardless of what it sits on) rather than a single stylistic trend (Figure 4 shows four representative examples). We have not tested this explanation against each institution's declared brand colours, and treat it as a hypothesis for future work rather than a finding.

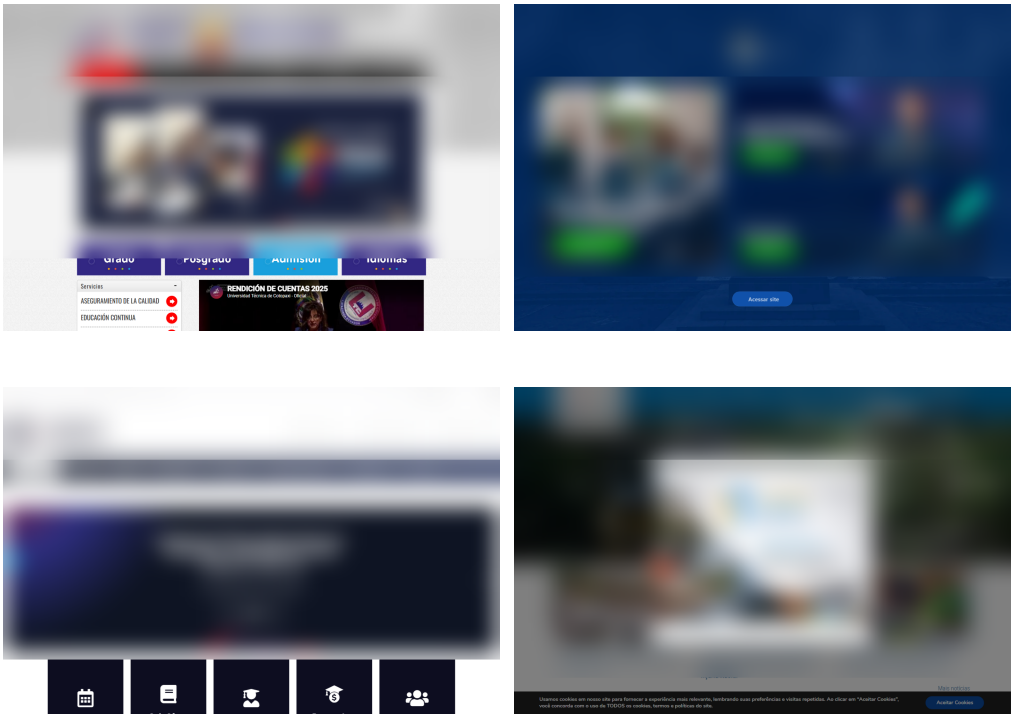


Figure 4: Homepages among the highest AA-failure rates in the sample, illustrating the brand-palette pattern described above: saturated institutional colours used directly as text or overlay colour rather than as accents. Captures show only public institutional cover content, not personal data.

A note on the WCAG contrast formula itself, relevant to interpreting the residual variance our models leave unexplained: WCAG's ratio is based on relative luminance, not on a perceptually uniform distance measure. Two pairs with identical WCAG ratios can differ substantially in perceived contrast, which is part of why our CIELAB-space clustering was necessary in the first place: the WCAG ratio alone cannot distinguish the qualitatively different failure patterns we report.

6 Limitations

This study has several limitations. First, we crawled homepages only, not full sites; a homepage that passes may sit atop a site with far worse compliance on interior pages (or vice versa). Second, we measured only colour contrast, which is one of many WCAG success criteria; a chromatically compliant site can still be inaccessible on other dimensions (keyboard navigation, alt text, semantic structure). Third, the crawl captures a single point in time (August 2026); institutional websites are redesigned periodically, so these figures are not stable indefinitely. Fourth, the source domain list had no entries for Bolivia or Venezuela, so the 17-country sample does not cover every country in the region. Fifth, and most consequential for how the country results should be read: colour pairs are clustered within sites (a single homepage contributes many pairs sharing one design system), so the pair-level χ^2 and logistic-regression tests overstate independent sample size and their p-values are optimistic; the site-level check we report alongside them (256 sites, $p = 0.085$) is not significant, so we treat the country pattern as suggestive rather than confirmed and do not rely on individual per-country coefficients. Sixth,

we used country as a proxy for institution type because a reliable public/private classification was not available across the source dataset; a finer-grained institutional variable, or a larger site-level sample powered to detect a modest effect, might resolve the country question more conclusively than either test here does. Seventh, our background-colour resolution walks the DOM ancestor chain for a non-transparent `background-color` and does not inspect `background-image`; a text element rendered over a photographic or gradient background resolves, incorrectly, against whatever solid colour sits further up the ancestor chain (typically page white). This is a plausible contributor to the single most frequent failing pair in our data (white text on a white background, 2,379 occurrences): at least part of that count is likely mislabelled text-over-image rather than genuinely invisible text, and we cannot currently distinguish the two cases from the extracted data alone. To bound how much this concern could be inflating our headline figures, we recomputed both under the most conservative reading, treating every single one of the 2,379 white-on-white pairs as an artefact and excluding them entirely: the pair-level failure rate drops from 21.6% to 16.8%, and the site-level share with at least one critical failure drops from 93.5% to 90.7% (only 7 of 259 sites, 2.7%, have white-on-white as their *only* recorded failure). Since at least some of these 2,379 pairs are surely genuine invisible-text bugs rather than background-image artefacts, this exclusion is an upper bound on the artefact's impact, not a corrected estimate; even so, both headline figures remain qualitatively unchanged. This also means our comparison to WebAIM's global figure (Results) uses a different detection instrument on a different population and should be read as an order-of-magnitude reference, not a controlled comparison. Finally, the moderate correlation with `axe-core` reflects genuine methodological differences between contrast-ratio counting and axe-core's rule engine rather than a validity problem with either approach, but it means the two should be read as complementary rather than interchangeable estimates.

7 Conclusion

A homepage-level survey of 259 Latin American university websites finds that WCAG AA contrast failures are the norm rather than the exception (93.5% of sites affected, 21.6% of all measured pairs); a country effect is visible at the pair level but is not confirmed once the actual unit of analysis (site) and the clustering of pairs within a site are accounted for, and we report it as an open question rather than a settled result. The colours involved in these failures are consistent with our initial hypothesis: unsupervised clustering in the full CIELAB $L^*a^*b^*$ space shows that near-neutral, low-chroma "light grey on white" patterns account for 80.0% of failing pairs, with four smaller chromatic clusters making up the remainder. This suggests that accessibility guidance for this region can prioritise the low-contrast neutral-text pattern as the primary target, while treating brand-palette contrast checking as a secondary recommendation for the smaller residual of saturated-colour failures.

Data Availability

The domain list, crawl logs, extracted colour pairs, and analysis code are permanently archived and openly accessible in Zenodo at `https://doi.org/10.5281/zenodo.22907769` and GitHub at `https://github.com/JuanAvilesP3/jged-contraste-cromatico`.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Use of Generative AI

Generative AI tools (large language models) were used during this work to assist with drafting and editing text, writing and debugging analysis scripts, LaTeX formatting, and pre-submission critical review. All

research questions, study design, data collection, executed analyses, reported results, and interpretations were defined, run, and verified by the authors, who take full responsibility for the content of this article. No figure was produced with image-generation tools.

Author Contributions

Juan Pablo Aviles-Esparza: Conceptualization, Methodology, Software, Writing - Original Draft. **Italo Javier Tenempaguay-Granizo:** Data Curation, Formal Analysis, Validation, Writing - Review & Editing. **Isaac David Torres-Paredes:** Supervision, Project Administration, Resources, Writing - Review & Editing.

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